Lie theory for posets and cones: a gentle introduction

October 2026

Abstract

The free Lie algebra hides inside the partition lattice: the top homology of the lattice of set partitions of $[n]$ is, up to a sign twist, the multilinear part of the free Lie algebra on n letters. This note explains, with small pictures and examples, how that classical fact extends first to finite posets and then to cones of real hyperplane arrangements and of oriented matroids. It describes the fixed-point formula for the characters that appear, says what is known about when those characters vanish, compares the picture with six neighbouring theories, and says what the generalization does and does not buy. It is a summary. The proofs are in the notes files named in the footnotes.

How to read this note

Every stated result has a box in the margin and a footnote naming the notes file (and section) it comes from. The boxes mean:

- PROVEN / AUDITED: the source note proves it, and an independent audit, with its own code, re-derived the proof and found that it holds. These are the only results stated as *Theorems*. Some of them rest on a cited published theorem; the footnote says which.
- CLASSICAL (CITED): a published result, stated as background.
- COMPUTED: checked by computer on the cases listed, not proved.
- CONJECTURED and OPEN: what they say.

When we say a result was "not found in the literature", we mean that the searches recorded in the notes did not find it. That is not a proof of novelty.

1 The classical story: Lie brackets in the partition lattice

1.1 Lie_n

Take the free Lie algebra on letters $x_1, \dots, x_n$ and keep the part that uses each letter exactly once. Call it Lie_n. It is spanned by bracketings such as $[[x_2, x_3], x_1]$. The symmetric group S_n acts by permuting the letters, and $\dim \mathrm{Lie}_n = (n-1)!$: a basis is the left-normed brackets $[\cdots[[x_1, x_{\sigma(2)}], x_{\sigma(3)}], \dots, x_{\sigma(n)}]$.

There is a second, very concrete model. Let $\gamma = (1\,2\,\cdots\,n)$, let ζ be a primitive nth root of unity, and put

$$e_\zeta = \frac{1}{n}\sum_{j=0}^{n-1} \zeta^{-j}\gamma^j \in \mathbb{C}[S_n].$$

Think of a permutation as a word $w = w_1 w_2 \cdots w_n$. Right multiplication by γ rotates the word, and $w\gamma e_\zeta = \zeta\, we_\zeta$. So we_ζ depends, up to a root of unity, only on the necklace (rotation class) of w.

Classical fact 1 (Klyachko)**.** $\mathrm{Lie}_n \cong \mathbb{C}[S_n]e_\zeta \cong \mathrm{Ind}_{C_n}^{S_n}\zeta$ as S_n-modules.[1]

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1.2 The partition lattice

Let Π_n be the set partitions of $[n]$, ordered by refinement. The proper part $\overline{\Pi_n}$ removes the bottom (all singletons) and the top (one block). Figure 1 shows Π_3: its proper part is three points, so its reduced homology is $\widetilde{H}_0$, of dimension $2 = (3-1)!$.

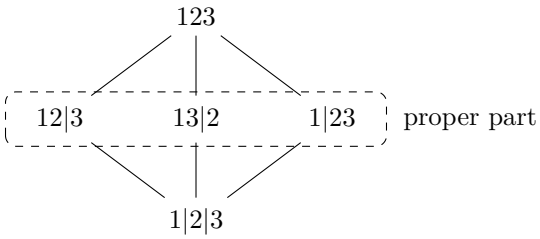


Figure 1: Π_3. The proper part is three points, a wedge of two 0-spheres.

Classical fact 2 (Joyal, Klyachko, Stanley; Hanlon)**.** $\overline{\Pi_n}$ has the homotopy type of a wedge of $(n-1)!$ spheres of dimension $n-3$, and

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$$\widetilde{H}_{n-3}(\overline{\Pi_n};\mathbb{C}) \cong \mathrm{sgn} \otimes \mathrm{Lie}_n \qquad \text{as } S_n\text{-modules.}$$

Hanlon and Stanley computed the character through fixed points: for g with k cycles all of length d, $\mu(\Pi_n^g) = (-1)^{k-1}\mu(d)\, d^{k-1}(k-1)!$, and $\mu(\Pi_n^g) = 0$ for every other $g \neq 1$.[2]

Here Π_n^g is the sublattice of partitions fixed by g, and μ is its Möbius number $\mu(\hat{0},\hat{1})$. The reason fixed points appear is the first of the three ideas this note is organized around.

Idea 1 (characters are Möbius numbers of fixed lattices). If the homology of a poset is concentrated in one degree, the Hopf trace formula and P. Hall's theorem turn the trace of a symmetry g on that homology into $\pm\mu$ of the subposet of g-fixed elements. Computing a character becomes computing a Möbius number.

2 Posets

2.1 Order-preserving partitions and linear extensions

Now replace the n unrelated letters by a finite poset P on $[n]$. A set partition π is *order-preserving* if collapsing each block to a point leaves a directed graph without cycles. These partitions form a

[1] As cited in notes/poset_jks_research_note.tex, §3.1 and notes/poset_jks_theorem.tex, Cor. 3.16(a).

[2] notes/poset_jks_theorem.tex, Cor. 3.16(a); Hanlon's method as read in notes/novelty_search_2.md, §2.1.

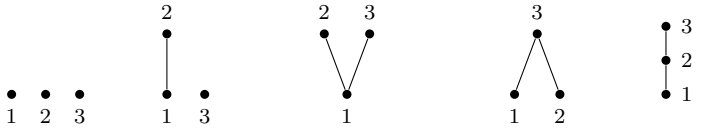
lattice $\mathcal{O}(P)$. For the antichain nothing is forbidden and $\mathcal{O}(P) = \Pi_n$. For the chain $1 < 2 < 3$ the partition $13|2$ is forbidden, because $1 < 2 < 3$ would become a cycle between the blocks $\{1, 3\}$ and $\{2\}$.

A *linear extension* of P is a word w in which i comes before j whenever $i <_P j$. Write $\mathrm{Lin}(P)$ for the set of them, and $e_C(P)$ for the number of rotation classes of words that contain at least one linear extension. If P is connected, no two linear extensions are rotations of each other, so $e_C(P) = |\mathrm{Lin}(P)|$.

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Classical fact 3 (Jenča–Sarkoci)**.** For $n \geq 3$, the proper part of $\mathcal{O}(P)$ is homotopy equivalent to a wedge of $e_C(P)$ spheres of dimension $n - 3$.[3]

Figure 2 lists the five posets on three points.



	antichain	$1 < 2,\ 3$	$1 < 2,\ 1 < 3$	$1 < 3,\ 2 < 3$	chain
$\lvert\mathrm{Lin}(P)\rvert$	6	3	2	2	1
$\lvert\mathcal{O}(P)\rvert$	5	5	5	5	4
$e_C(P) = \dim \widetilde{H}_0$	2	2	2	2	1

Figure 2: The posets on three points, by direct enumeration. For the second poset the three linear extensions $123, 312, 132$ form two rotation classes, $\{123, 312\}$ and $\{132\}$. For the chain, $\mathcal{O}(P)$ is the Boolean lattice $\{1|2|3,\ 12|3,\ 1|23,\ 123\}$.

2.2 The poset Lie module

Define

$$\mathsf{Lie}(P) = \mathrm{span}\{we_\zeta : w \in \mathrm{Lin}(P)\} \subseteq \mathbb{C}[S_n].$$

Its dimension is $e_C(P)$, and the automorphism group $\mathrm{Aut}(P)$ acts on it by relabelling letters. For the antichain it is all of $\mathbb{C}[S_n]e_\zeta$, which is Lie_n. Let $H_P = \widetilde{H}_{n-3}(\overline{\mathcal{O}(P)}; \mathbb{C})$.

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Theorem 1 (poset JKS)**.** *Let P be a finite poset on $n \geq 3$ points. Then $H_P \cong \mathrm{sgn} \otimes \mathsf{Lie}(P)$ as $\mathrm{Aut}(P)$-modules. Equivalently, for every $g \in \mathrm{Aut}(P)$,*

$$(-1)^{n-3}\mu(\mathcal{O}(P)^g) = \mathrm{sgn}(g) \sum_{C\,:\,gC=C} \zeta^{r_g(C)},$$

where C runs over the rotation classes meeting $\mathrm{Lin}(P)$ that g maps to themselves, and $r_g(C)$ is the rotation by which g acts on C. Moreover both sides equal $\mathrm{sgn}(g)\,\mu(o) \sum_{x \in F(P,g)} e_C(P|_{B_0(x)})$, where o is the order of g, $\mu(o)$ is the number-theoretic Möbius function, and $F(P, g)$ is the set of partitions into o blocks, each a union of connected components, that g permutes cyclically ($B_0(x)$ is any block).[4]

[3] Cited as input (A) in notes/poset_jks_theorem.tex, §2; the statement was checked against the source in notes/citation_check.md, §2.

[4] notes/poset_jks_theorem.tex, §3, Theorem 3.1. Two independent proofs, both audited in notes/audit_character_identity_proof.md ("proof holds"). The proof uses Jenča–Sarkoci, P. Hall's theorem and the Hopf trace formula, all cited. The isomorphism is abstract (equal characters); an explicit choice-free isomorphism is in §6 of the same file.

For the antichain this is the classical JKS theorem. The right-hand side of the character identity has a plain meaning: it is a sum over the rotation classes of linear extensions that g fixes, each weighted by a root of unity. At $g = 1$ it is the count $e_C(P)$, and the identity becomes $\mu(\mathcal{O}(P)) = (-1)^{n-1} e_C(P)$. That count also has a proof with no topology at all: one application of Weisner's theorem and Jenča–Sarkoci's recursion for e_C.[5]

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Corollary 2. *If P is connected and $n \geq 3$, then $\mu(\mathcal{O}(P)^g) = 0$ for every $g \neq 1$, and H_P is a free $\mathrm{Aut}(P)$-module of rank $|\mathrm{Lin}(P)|/|\mathrm{Aut}(P)|$.*[6]

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So roots of unity appear only when g permutes connected components freely.

Example 1 (a fixed-point computation)**.** Let P be two disjoint chains $1 < 2$ and $3 < 4$, and let $g = (13)(24)$ swap them. The linear extensions are the six shuffles of 12 and 34. The words 1234 and 3412 are rotations of each other, and the other four lie in four further classes, so $e_C(P) = 5$ and $\dim H_P = 5$.

Lie side. Only the class $\{1234, 3412\}$ is fixed by g: g sends 1234 to 3412, which is 1234 rotated by two places. With $\zeta = i$ the phase is $\zeta^2 = -1$. So the trace of g on $\mathsf{Lie}(P)$ is -1.

Möbius side. The g-fixed order-preserving partitions are drawn in Figure 3. Counting upwards, μ is -1 on the three atoms, $+1$ on 13|24 and $+1$ on the top. So $(-1)^{4-3}\mu(\mathcal{O}(P)^g) = -1$, and $\mathrm{sgn}(g) = +1$. The two sides agree. The closed formula gives the same number: $o = 2$, $\mu(2) = -1$, $F(P, g) = \{12|34\}$, and e_C of a 2-chain is 1.

Why it works (Weisner). The orbit partition of g is $O_g = 13|24$. Weisner's theorem says that the Möbius numbers $\mu(\hat{0}, x)$ of the x with $x \vee O_g = \hat{1}$ sum to zero. Here those x are 12|34 and $\hat{1}$, so $\mu(\hat{0}, \hat{1}) = -\mu(\hat{0}, 12|34) = 1$. The general proof is this step plus a Möbius inversion.[7]

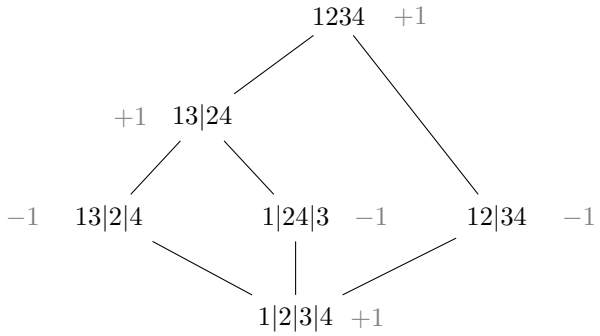


Figure 3: The lattice $\mathcal{O}(P)^g$ for $P = \{1 < 2\} \sqcup \{3 < 4\}$ and $g = (13)(24)$, with $\mu(\hat{0}, x)$ in grey.

Remark. The first proposed route to Theorem 1 was to show that one fixed classical JKS isomorphism maps H_P onto $\mathsf{Lie}(P)$. That is false already for a chain and its reverse: they have the same $\mathcal{O}$, but different spans $\mathbb{C}\, we_\zeta$.[8] The correct restricted statement replaces $\mathbb{C}[S_n]e_\zeta$ by $\mathbb{C}[S_n]$ modulo shuffles. That leads to the second idea.

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[5] notes/mu_op_elementary.md, Theorem A (Proof W), audited in notes/audit_mu_op_elementary.md: both proofs hold and are homology-free.

[6] notes/poset_jks_theorem.tex, Cor. 3.15; audited in notes/audit_character_identity_proof.md, II7.

[7] notes/poset_jks_theorem.tex, §3.2, Prop. 3.11 and Lemma 3.8 (Weisner). The computation in this example was done by hand and by direct enumeration.

[8] notes/poset_jks_theorem.tex, §4, Prop. 4.2; audited in notes/audit_literature_novelty_jks_maps.md.

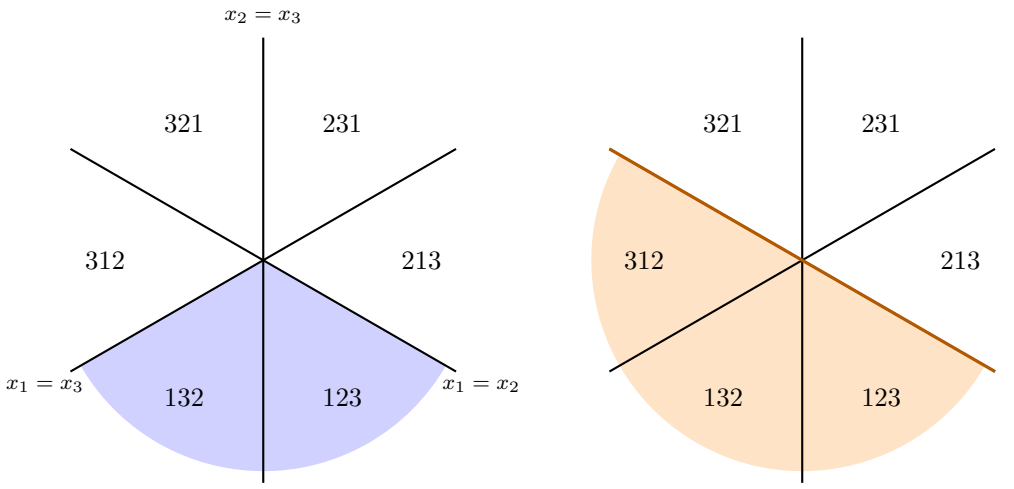
3 Geometry: chambers, cones and lunes

3.1 The braid arrangement

The braid arrangement is the set of hyperplanes $x_i = x_j$ in $\mathbb{R}^n$. Its chambers (the regions where all coordinates differ) are the permutations: the chamber of w is $x_{w_1} < x_{w_2} < \cdots < x_{w_n}$. A poset cuts out the *order cone*

$$C_P = \{x \in \mathbb{R}^n : x_i \le x_j \text{ whenever } i <_P j\},$$

and the chambers inside C_P are exactly the linear extensions of P. Figure 4 shows $n = 3$, after quotienting by the line $x_1 = x_2 = x_3$: three lines, six chambers.



(a) $P = \{1 < 2,\ 1 < 3\}$: a pointed cone (b) $P = \{1 < 2\}$: a half-plane, a top-lune

Figure 4: The braid arrangement A_2. (a) The order cone of $1 < 2,\ 1 < 3$ contains the chambers 123 and 132. Its faces span three lines, so its support lattice has three middle elements and $m(V) = 2$. (b) The order cone of $1 < 2$ is a half-plane containing $123, 132, 312$. It contains the line $x_1 = x_2$ (its lineality space), and it is itself a top-lune.

Faces of the arrangement inside C_P correspond to order-preserving maps $P \to [k]$ onto chains, and the linear span of a face is recorded by the partition into fibres. In this way the lattice of spans of faces of C_P is $\mathcal{O}(P)$, turned upside down.[9]

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3.2 Any arrangement, any cone

Let $\mathcal{A}$ be a finite central arrangement of hyperplanes in $\mathbb{R}^d$, essential (the hyperplanes meet only in 0). A *top-cone* V is a nonempty set of chambers obtained by choosing a side of some of the hyperplanes. Its *support lattice* $\Lambda(V)$ is the set of subspaces spanned by faces of $\mathcal{A}$ that lie in the closed cone $|V|$, ordered by inclusion. Put $m(V) = (-1)^d \mu(\Lambda(V))$. The *lineality space* $\operatorname{lin} V = |V| \cap -|V|$ is the largest subspace inside the cone. The cone is *pointed* if $\operatorname{lin} V = 0$.

Theorem 3 (cone wedge theorem)**.** *For every top-cone V of a finite real central essential arrange-*

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[9] notes/cone_lune_lie_theory.md, §2, Prop. 2.1 (audited in notes/audit_cone_lune_lie_theory.md), and notes/general_theory.md, §1.3, Cor. 1.4 (audited in notes/audit_general_theory_cones.md).

ment of rank d, the proper part of $\Lambda(V)$ is homotopy equivalent to a wedge of $m(V)$ spheres of dimension $d-2$.[10]

For the braid arrangement and $V = C_P$ this is the Jenča–Sarkoci theorem, with a new proof, and $m(C_P) = e_C(P)$. For $V =$ all chambers it is Folkman's theorem for geometric lattices.

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Theorem 4 (Zaslavsky for cones)**.** *$\#V = \sum_X m(V_X)$, where X runs over the flats contained in $\operatorname{lin} V$ and V_X is the image of V in $\mathbb{R}^d/X$. If V is pointed, $\#V = m(V)$.*[11]

In Figure 4(b), $\#V = 3$, $m(V) = 2$, and the line $x_1 = x_2$ contributes 1. In the braid case the theorem reads $|\operatorname{Lin}(P)| = \sum_\sigma \prod_{B\in\sigma} e_C(P|_B)$, a sum over partitions σ of the components of P.

3.3 Lie elements are what lunes cannot see

A *top-lune* is the set of chambers lying on prescribed sides of all the hyperplanes through some fixed flat. In the braid arrangement the top-lunes are the sets of shuffles of a few words, for example $\{123, 132, 312\}$, the shuffles of 12 and 3. Let Γ be the space spanned by the chambers, and let $\mathrm{Lune} \subseteq \Gamma$ be spanned by the sums over top-lunes with more than one chamber. The Lie elements of the arrangement, in the sense of Aguiar–Mahajan, are the vectors orthogonal to Lune. For the braid arrangement Lune is the span of the proper shuffles, and its orthogonal is Lie_n (Ree).[12]

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Idea 2 (chambers modulo lunes). The Lie module of a cone is the span of its chambers, modulo the lunes. The top homology of the cone's support lattice is that module, twisted by the determinant.

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Theorem 5 (cone JKS)**.** *Let V be a top-cone of a finite real central essential arrangement in $\mathbb{R}^d$, $d \geq 2$, and let G be a finite group of linear maps preserving $\mathcal{A}$. Write Γ_V for the span of the chambers of V. Then*

$$\widetilde{H}_{d-2}(\overline{\Lambda(V)};\mathbb{C}) \cong \det \otimes (\Gamma_V + \mathrm{Lune})/\mathrm{Lune}$$

as modules over the stabilizer of V in G. In particular $m(V)$ equals $\dim \Gamma_V/(\Gamma_V \cap \mathrm{Lune})$.[13]

For C_P the determinant is sgn, and Theorem 5 says $H_P \otimes \mathrm{sgn} \cong \mathbb{C}[\operatorname{Lin} P]/(\text{lunes inside } C_P)$. That is the poset theorem in a form that names an explicit basis: the splitting cycles of the linear extensions.[14] In Figure 4(b), three chambers modulo one lune leave a two-dimensional space, and $m(V) = 2$. A pointed cone contains no lune with more than one chamber, and its chambers are independent modulo lunes.

[10] notes/general_theory.md, §1.1, Theorem 1; audited in notes/audit_general_theory_cones.md (holds; the cited fibre theorem of Björner–Wachs–Welker can be removed, Prop. A there). We did not find it in the literature (notes/novelty_general_theory.md, §2).

[11] notes/general_theory.md, §1.2, Theorem 2; audited in notes/audit_general_theory_cones.md. It is a short specialization of Zaslavsky's 1977 dissection theorem (notes/novelty_general_theory.md, §3).

[12] Aguiar–Mahajan, *Topics in Hyperplane Arrangements*, Lemmas 10.7–10.8, as read in notes/literature_novelty_jks_maps.md, §4.2, and notes/cone_lune_lie_theory.md, §1.

[13] Proved as Theorem C in notes/audit_general_theory_cones.md, §C, given a theorem of Brown on left regular bands (cited, read in the source); restated as Theorem G* in notes/general_theory.md, §1.5. Independently audited in notes/audit_theorem_c.md (holds with fixes to proofs; no conclusion changes). For $V =$ all chambers it is known (Aguiar–Mahajan, *Topics*, Thm. 14.32); for proper cones we did not find it (notes/novelty_general_theory.md, §5).

[14] notes/poset_jks_theorem.tex, §5, Theorem 5.3(d), which is proved there given a cited theorem of Wachs; by Theorem 5 it also holds given Brown (notes/general_theory.md, §1.5).

3.4 Oriented matroids

Nothing above needs coordinates. An oriented matroid keeps only the sign vectors of the faces, and some oriented matroids come from no real arrangement. The role of a top-cone is played by a T-convex set of topes.

Theorem 6. *For every simple oriented matroid of rank r and every nonempty T-convex set V of topes: the proper part of $\Lambda(V)$ is a wedge of $m(V)$ spheres of dimension $r-2$; the Zaslavsky count holds; and the top homology is* $\det \otimes (\Gamma_V + \mathrm{Lune})/\mathrm{Lune}$ *as a module over the stabilizer of V, given Brown. Realizability is not used.*[15]

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4 Characters: the fixed-point formula

Let g be a linear map of finite order with $g\mathcal{A} = \mathcal{A}$ and $gV = V$, for instance an automorphism of P acting on C_P. By Idea 1 its character on the top homology is

$$\chi_V(g) = (-1)^d \det(g)\, \mu(\Lambda(V)^g),$$

where $\Lambda(V)^g$ is the lattice of g-stable spans. In the braid case this is the left side of Theorem 1, multiplied by $\mathrm{sgn}(g)$.

Theorem 7 (equivariant cone Zaslavsky)**.** *The number of chambers of V fixed by g is*

$$\#V^g = \sum_X \chi_{V_X}(\bar{g}),$$

the sum over the g-stable flats $X \subseteq \operatorname{lin} V$, where $\bar{g}$ is the map induced on $\mathbb{R}^d/X$. Inverting this over the lattice of g-stable flats gives a closed formula for $\chi_V(g)$ in terms of chambers fixed by g, which can be counted on the fixed space $\mathrm{Fix}(g)$.[16]

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At $g = 1$ this is the cone Zaslavsky theorem. On a pointed cone it says $\chi_V(g) = \#V^g$. If $\mathcal{A}$ is the reflection arrangement of a reflection group W and $1 \neq g \in W$, then g fixes no chamber, so the top homology of a pointed cone is a free module over the stabilizer of V in W.[17] This is the analogue of the connected-poset corollary. For the braid arrangement the formula of Theorem 7 agrees with the closed formula of Theorem 1.

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5 When does the character vanish?

Idea 3 (transverse flats). Whether $\chi_V(g)$ can be nonzero is decided by the g-stable flats X that meet $\mathrm{Fix}(g)$ only in 0 and are transverse to it (no hyperplane contains $X + \mathrm{Fix}(g)$), and by the Möbius numbers of their fixed lattices.

Write $Z^*(V, g)$ for the g-stable flats $X \subseteq \operatorname{lin} V$ with $X \cap \mathrm{Fix}(g) = 0$ such that no hyperplane of $\mathcal{A}$ contains $X + \mathrm{Fix}(g)$, and $Z^*(g)$ for the same set with $V =$ all chambers.

Theorem 8 (Formula Z and the some-cone criterion)**.**

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[15] notes/oriented_matroid_cones.md, Verdict table, Theorems 1-OM, 2-OM and C-OM; audited in notes/audit_oriented_matroid_cones.md: every claim holds, with one wrong attribution fixed. For $r \geq 5$ the homotopy form uses shellability of the tope complex (cited).

[16] notes/fixed_point_formula.md, Verdict table, rows EZ and CF; audited in notes/audit_fixed_point_formula.md (holds, with fixes). For $V =$ all chambers both are known in substance (Moseley; Sundaram–Welker); for proper cones we did not find them (notes/novelty_fixed_point_formula.md, §§2–3).

[17] notes/fixed_point_formula.md, Cor. P; audited in notes/audit_fixed_point_formula.md.

$$\chi_V(g) = \sum_{X \in Z^*(V,g)} \mu_{L(\mathcal{A})^g}(0, X)\, m_X(V), \qquad m_X(V) \geq 1,$$

where $L(\mathcal{A})^g$ is the lattice of g-stable flats and $m_X(V)$ is the multiplicity of a smaller cone inside $\mathrm{Fix}(g)$. *Consequently $\chi_V(g) \neq 0$ for some g-stable top-cone V if and only if some $X \in Z^*(g)$ has $\mu(0, X) \neq 0$, and then a top-lune witnesses it.*[18]

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Theorem 9 (type A, all of $\pm S_n$)**.** *On the braid arrangement in $\mathbb{R}^n$ ($n \geq 3$), let $g = \pm w$ with $w \in S_n$. Some g-stable cone has $\chi \neq 0$ if and only if $\langle g \rangle$ acts freely on $\{\pm e_1, \ldots, \pm e_n\}$ and $\mu(\mathrm{ord}\, g) \neq 0$. For $+w$ this is the classical rule "all cycles of one length o, with $\mu(o) \neq 0$". For $-w$ it says that the cycle lengths of w lie in $\{m, 2m\}$ for one odd squarefree m.*[19]

The map $-w$ preserves an order cone C_P exactly when w is an anti-automorphism of P (it reverses every relation), so the second half is about anti-automorphisms of posets. Since $\mathcal{O}(P) = \mathcal{O}(P^{\mathrm{op}})$, they act on $\mathcal{O}(P)$ too.[20] Already for $n \leq 7$ there are nine classes $-w$ that are nonzero on some cone but zero on all chambers, for example $-(2, 1)$.[21] The rule for $-w$ is invisible if one looks only at all chambers.

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Per cone. Formula Z can cancel: nonzero terms of opposite sign can add to zero. In type A this is completely understood.

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Theorem 10 (type A, every cone)**.** *Let τ be an anti-automorphism of P ($n \geq 3$). Call a point odd if it lies on an odd cycle of τ. Then $\chi_{C_P}(-\tau) \neq 0$ if and only if*

(1) the cycle lengths of τ lie in $\{m, 2m\}$ for one odd squarefree m;

(2) the $\langle \tau \rangle$-orbit of every component K of P has length m or $2m$;

(3) for every component K, the set $K \cup \tau^m K$ contains at most one odd point;

(4) if τ has both odd and even cycles, some odd cycle of τ does not consist of isolated points of P.

There is a positive closed formula for $|\chi_{C_P}(-\tau)|$. With Theorem 1 for $+w$, this decides vanishing for every element of $\pm S_n$ on every cone of the braid arrangement.[22]

One ingredient is a small fact about linear extensions: an anti-automorphism t of a poset has a linear extension w with $t(w_i) = w_{n+1-i}$ for all i (a "t-palindromic" extension) if and only if $t^2 = 1$ and t has at most one fixed point.[23] These palindromic extensions are the chambers that $-t$ fixes.

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What is open.

- An exact per-cone vanishing criterion beyond type A: for B_n ($n \geq 5$), D_n, H_3, H_4, F_4 and general arrangements. Off the Coxeter examples, cancellation is common.[24]

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[18] notes/vanishing_criterion.md, Verdict table, rows Z and S; audited twice, in notes/audit_vanishing_criterion.md and notes/audit2_vanishing_criterion.md (both: holds). We did not find either statement for proper cones (notes/prior_art_vanishing_criterion.md, §§2–3).

[19] notes/vanishing_criterion.md, row A and §4; audited in both audits named above.

[20] notes/per_cone_vanishing.md, §3; notes/poset_jks_theorem.tex, Lemma 4.1.

[21] notes/prior_art_vanishing_criterion.md, §4.

[22] notes/per_cone_vanishing.md, §3, Theorem TA and row TA±. Audited in notes/audit_per_cone_vanishing.md: "holds with fixes"; the fixes are to proofs and do not change a result. (The status line inside the source note was written before that audit.)

[23] notes/per_cone_vanishing.md, §2, Lemma Pal; audited in notes/audit_per_cone_vanishing.md (holds; checked on all 6,847 anti-automorphisms with $n \leq 7$).

[24] Rows PC′ and PC″ of notes/vanishing_criterion.md, and rows B, D, H, F of notes/per_cone_vanishing.md.

- The Crapo-type converse. By Crapo's complementation theorem, if the interval $[0, X]$ of the fixed lattice has an element with no complement, then $\mu(0, X) = 0$. On all chambers, the converse held for all 191 flats $X \in Z^*(g)$ tested.[25] On cones, the conjecture is: $\chi_V(g) = 0$ if and only if $\Lambda(V)^g$ has an element with no complement. No counterexample was found on 18,750 pairs (V, g).[26] CONJECTURED
- In type A, whether $\chi_V(g) = 0$ depends only on $(\lin V, g)$.[27] PROVEN AUDITED
- In general it does not: generic arrangements of four planes in $\mathbb{R}^3$ already give counterexamples.[28] COMPUTED

6 Neighbouring theories

Each paragraph names the analogy and where it stops. Several primary sources could not be opened, and the notes record them as *not accessible*; statements about them below come from secondary accounts.

Orlik–Solomon algebras and Lehrer–Solomon. The Orlik–Solomon algebra is the cohomology of the complement of the complexified arrangement. Lehrer–Solomon describe the S_n-action on it as a sum of modules induced from centralizers. On all chambers, the cone character χ is the top Orlik–Solomon character twisted by det (Orlik–Solomon 1980, Thm. 4.8).[29] The analogy stops at cones. That theory concerns the whole complement, so it sees only all chambers. The cone theory decomposes the space spanned by chambers instead. For all chambers that decomposition matches the cohomology of a different space, Moseley's complement $M_3(\mathcal{A})$ of the codimension-three subspaces, and it differs from the Orlik–Solomon one by a determinant twist on each summand. Nothing in those accounts mentions fixed chambers, cones or lineality spaces.

Lie characters of Coxeter groups. For a reflection group W, the Lie module of the whole arrangement is a W-module, and Aguiar–Mahajan prove a JKS theorem for every arrangement, with the twist carried by an orientation space. This is the all-chambers case of Theorem 5.[30] Bishop, Douglass, Pfeiffer and Röhrle give a support condition for the top Orlik–Solomon component through cuspidal classes, for groups of rank at most 8 and for all chambers only. The cone theory adds proper cones and symmetries outside W, such as $-w$ in type A. The type-A support rule does not generalize in the obvious way: Springer regularity is neither necessary nor sufficient for being nonzero on some cone. The Coxeter element of B_3 vanishes on every cone, and a reflection of B_2 does not.[31] PROVEN AUDITED

Aguiar–Mahajan: Hopf monoids, cones and lunes. The vocabulary here (Tits product, top-lunes, Lie elements as Lune$^\perp$) is Aguiar–Mahajan's. Their books contain no theory attached to a single cone, and one exercise needed correcting: an element is Lie iff its coefficient sum vanishes

[25] notes/vanishing_criterion.md, row CR.

[26] notes/per_cone_vanishing.md, row CR$_V$ and §4.

[27] notes/per_cone_vanishing.md, row TA-L (Cor. 3.5); audited in notes/audit_per_cone_vanishing.md.

[28] notes/per_cone_vanishing.md, row L′ and §4.

[29] notes/prior_art_vanishing_criterion.md, Verdict table and §0.3; notes/novelty_fixed_point_formula.md, §4. Lehrer 1987 and Orlik–Solomon 1980 were read only through secondary accounts.

[30] notes/novelty_general_theory.md, §5.

[31] notes/vanishing_criterion.md, row R; proven by hand given the some-cone criterion, audited in notes/audit_vanishing_criterion.md.

on every *non-pointed* top-cone, not every non-singleton one.[32] The natural first guess, the left regular band of faces of the cone acting on its own chambers, gives zero for the chain, and on all proper order cones tested. The right object is the ambient Lie module restricted to the cone.[33] On the Hopf side, a Hopf monoid of pairs (poset, linear extension), with coproduct by restriction to unions of components, has primitives $\cong \mathsf{Lie}(P)$ in each degree P and a poset-graded Lie bracket, and a local PBW theorem holds.[34] The bracket is not induced by concatenation in the e_ζ model, and a version for general arrangements (via Aguiar–Mahajan's bimonoids) has not been worked out.

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Whitehouse modules and Lie in type B. The Whitehouse module is a module for the larger group S_{n+1} whose restriction to S_n is Lie_n (classical background, not discussed in the notes); type-B analogues of the free Lie algebra (Bergeron; Gottlieb–Wachs; Aguiar–Mahajan §14.9) are modules for the hyperoctahedral group on the B_n arrangement. The analogy is "extra symmetry": the cone theory sees the group generated by $\mathrm{Aut}(P)$ and the anti-automorphisms acting as $-\tau$, which is a subgroup of $\pm S_n$, and the $-w$ rule above is a statement about it. It stops at the arrangement: type-B Lie characters are traces on the B_n arrangement, while $-w$ here acts on cones of the A_{n-1} arrangement. Whether some type-B Lie character has the same support as the $\{m, 2m\}$ rule was not checked, and a search for Whitehouse modules together with linear extensions found nothing relevant.[35] Whether H_P carries a hidden action of a group larger than $\mathrm{Aut}(P)$, in the spirit of the Whitehouse module, is not studied in the notes.

Hanlon's fixed-point Möbius computations. Hanlon computed $\mu(\Pi_n^g)$ using the *hinge*, the partition of $[n]$ into cycles of g, and its complements. The proof of Theorem 1 applies Weisner's theorem at the same element (the orbit partition O_g), and for the antichain its set $F(P, g)$ is exactly Hanlon's complements. On all chambers of the braid arrangement, Formula Z is Hanlon's count of complements in substance.[36] The analogy stops at the lattice. Hanlon's method uses supersolvability and geometric lattices, and $\mathcal{O}(P)^g$ is neither in general. Weisner plus Möbius inversion needs neither. We did not find the poset formula of Theorem 1 in the literature. Whether it follows from Baclawski–Björner's work on fixed points and complements is open, because that paper could not be read.

Oriented matroids. As explained above, the wedge theorem, the Zaslavsky count and the cone JKS theorem hold for T-convex sets of topes of any simple oriented matroid. They do not need realizability, because Brown's theory of left regular bands is stated for any finite left regular band.[37] The notes state the fixed-point formula and the vanishing results only for real arrangements with linear symmetries, and the sections on more general objects (complexes of oriented matroids) have not been audited.

[32] notes/cone_lune_lie_theory.md, §7, audited in notes/audit_cone_lune_lie_theory.md; notes/literature_novelty_jks_maps.md, §4.3.

[33] notes/cone_lune_lie_theory.md, Verdict row 3; proven for the chain, computed for $n \leq 5$.

[34] notes/general_theory.md, §2, Theorems 6–8; audited in notes/audit_general_theory_hopf.md (hold, with wording fixes).

[35] notes/prior_art_vanishing_criterion.md, §4 and §0.3 (Bergeron 1991 not accessible); notes/literature_novelty_jks_maps.md, §1.4 (query log).

[36] notes/novelty_search_2.md, §§2.1–2.2; notes/prior_art_vanishing_criterion.md, §2.1.

[37] notes/audit_oriented_matroid_cones.md, §1.

7 Why it might matter

7.1 The motivation: moonshine

Monstrous moonshine begins with a numerical coincidence. The coefficient 196884 of the modular j-function is $1 + 196883$, the sum of the two smallest dimensions of irreducible representations of the Monster group. McKay, Thompson, Conway and Norton conjectured that each coefficient of j is the dimension of a graded piece of a Monster module, and that replacing dimensions by traces of each group element gives other modular functions. Borcherds proved this. An important step was a Lie algebra with Monster symmetry whose denominator identity is a product formula of Witt type.

This program was started by asking whether something analogous could exist for the alternating groups. The research note that started it describes a chain of the same shape:[38]

$$\begin{aligned}&\text{a large combinatorial family} \;\to\; \text{equivariant homology}\\ &\quad\to\; \text{Lie characters} \;\to\; \text{Witt / plethystic-logarithm identities.}\end{aligned}$$

For posets, the last two arrows are now theorems. Sum $\mathsf{Lie}(P)$ over all labelled posets on $[n]$ to get an S_n-module V_n. Let a_n count naturally labelled posets, $A(x) = 1 + \sum_n a_n x^n$, and let $b_n = n[x^n] \log A(x)$.

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Theorem 11. *b_n counts the labelled posets having a rotation of $12\cdots n$ as a linear extension; $V_n \cong \mathrm{Ind}_{C_n}^{S_n}(\zeta \otimes \mathbb{C}[B_n])$ for that set B_n; and*

$$\mathrm{ch}\, V_n = \frac{1}{n}\sum_{d\mid n} \mu(d)\, b_{n/d}\, p_d^{n/d}, \qquad \sum_{n\geq 1} \mathrm{ch}\, V_n = \sum_{d\geq 1} \frac{\mu(d)}{d} \log A(p_d).$$

The global homology module is $\mathrm{sgn} \otimes V_n$.[39]

Compare Witt's formula $\mathrm{ch}\,\mathrm{Lie}_n = \frac{1}{n}\sum_{d\mid n}\mu(d)p_d^{n/d}$: the single generator is replaced by the series of naturally labelled posets. Restricted to even permutations the sign twist disappears, so on $\mathrm{Aut}(P) \cap A_n$ the homology H_P is simply $\mathsf{Lie}(P)$.

No modular forms have been produced. This is a motivation, not a result. The point of the general theory, from this angle, is a precondition: before looking for an extra grading whose traces could be modular, one needs a Lie theory for posets that is coherent, with modules, characters, functoriality and global generating functions that fit together. The results above supply the modules, the characters and the global generating function. Functoriality is only partial: an isomorphism natural along order extensions exists exactly on connected posets, and the e_ζ model carries no canonical bracket.[40] The search for such a grading has not begun.[41]

OPEN

7.2 A second measure: counting and structure of linear extensions

Does the theory help to count linear extensions, or to understand their structure? Here is where it touches them, in what the notes prove.

[38] notes/poset_jks_research_note.tex, section "Relation to the moonshine motivation".

[39] notes/poset_jks_theorem.tex, §7: Prop. 7.1, Cor. 7.2, Theorem 7.3 and Prop. 7.4(a)–(c); audited in notes/audit_cyclic_compatible_posets_bn.md and notes/audit_character_identity_proof.md. The sign-twisted generating function of Prop. 7.4(d) has been computed but has had no separate audit.

[40] notes/poset_jks_theorem.tex, §6, Theorem 6.8, and §9, items 1–2.

[41] notes/poset_jks_theorem.tex, §9, item 7.

- $e_C(P)$ counts the rotation classes of linear extensions, and $\mu(\mathcal{O}(P)) = (-1)^{n-1} e_C(P)$. A count becomes a Möbius function, and the proof needs no topology.
- The cone Zaslavsky theorem writes $|\operatorname{Lin}(P)|$ as a sum of products of e_C over partitions of the components of P.
- The character of $g \in \operatorname{Aut}(P)$ is a sum over the rotation classes of linear extensions fixed by g, weighted by roots of unity. The closed formula says this sum is $\mu(o)$ times a count of e_C's of blocks.
- For anti-automorphisms, the fixed chambers are the palindromic linear extensions, and Lemma Pal says exactly when they exist.

Question 1. Do these identities give anything new for counting $|\operatorname{Lin}(P)|$ itself, which is hard in general? The notes give no such application.

Question 2. The theory uses one cyclic action on words, rotation. Is there a link with other cyclic actions on linear extensions, such as promotion? This is not studied in the notes.

7.3 What the general theory buys over the poset theorem

- *Unification.* Classical JKS, Folkman, Zaslavsky, Jenča–Sarkoci, Hanlon's values of $\mu(\Pi_n^g)$ and the component formula become special cases of three statements: the wedge theorem, the cone JKS theorem, and the fixed-point formula.
- *Computations.* The fixed-point formula was checked on 7,200 pairs (cone, symmetry) in types A_2–A_5, B_2–B_4, D_4 and H_3.[42] The equivariant cone JKS theorem was checked on proper cones, and the determinant twist is visible there: without it the identity fails on 9 of the 33 cones of B_2.[43] COMPUTED
- *The vanishing question.* The poset theorem alone does not suggest it. The cone theory answers "nonzero on some cone" exactly, answers "nonzero on this cone" in type A, and reduces the rest to a precise Crapo-type conjecture.

What is still open. The exact per-cone vanishing criterion outside type A; the Crapo-type converses (conjectured); a canonical bracket or cooperad on the e_ζ model; a version of the Hopf-monoid picture for general arrangements; and the formalization below.

[42] notes/fixed_point_formula.md, Verdict table.
[43] notes/general_theory.md, Summary item 4.

8 Status of the Lean formalization

Lean status, as of 31 October 2026 (from lean/PLAN.md, §4.3). Lean 4 with Mathlib.

- **Sorry count: 1**, in MainTheorem.lean. It is `jencaSarkoci_homology`, the Jenča–Sarkoci theorem (a cited input), which is *not yet formalized*.
- **Proved without sorry**: the character identity of Theorem 1 and its closed formula, by a homology-free route (they use only the standard axioms); the Lie-side trace; P. Hall's theorem; the Hopf trace formula; the trace of g on H_P from the vanishing part of Jenča–Sarkoci; "equal characters imply isomorphic"; and Theorems S(a), S(b), Ψ and Cor. Ψ.1 (the explicit isomorphism $\mathsf{Lie}(P) \cong \mathbb{C}[\mathrm{Lin}\, P]/R_P$).
- The isomorphism $H_P \cong \mathrm{sgn} \otimes \mathsf{Lie}(P)$ (`main_iso`) is proved from `jencaSarkoci_homology`; it uses no other unproved statement.
- The cone, oriented-matroid, fixed-point and vanishing theory is not formalized.

Where to read more

notes/poset_jks_theorem.tex	the poset theorem, its proofs, the global module
notes/general_theory.md	cones: wedge, Zaslavsky, cone JKS, Hopf monoids
notes/oriented_matroid_cones.md	the oriented-matroid version
notes/fixed_point_formula.md	characters on cones
notes/vanishing_criterion.md	Formula Z, some-cone criterion, type A
notes/per_cone_vanishing.md	per-cone vanishing, Crapo, Conjecture CR_V
notes/prior_art_vanishing_criterion.md	what was searched and found
notes/novelty_*.md	the same, for the other results
notes/citation_check.md	cited statements checked against sources
lean/PLAN.md	the formalization